

Continuously Tunable Thin-Film Lithium Niobate Optical Delay Line with 293 ps Tuning Range

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We present a compact optical delay line (ODL) with wide-range continuous tunability on thin-film lithium niobate platform. The proposed device integrates an unbalanced Mach-Zehnder interferometer (MZI) architecture with dual tunable couplers, where each coupler comprises two 2×2 multimode interferometers and a MZI phase-tuning section. Experimental results demonstrate continuous delay tuning from 0 to 293 ps through synchronized control of coupling coefficients, corresponding to a 4 cm path difference between interferometer arms. The measured delay range exhibits excellent agreement with theoretical predictions derived from ODL waveguide parameters. This result addresses critical challenges in integrated photonic systems that require precise temporal control, particularly for applications in optical communications and quantum information processing, where a wide tuning range is paramount.

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The rapid advancement of optical communication, quantum information processing, and photonic computing has driven the development of high-performance integrated photonic devices.^[1–3] Among these, the optical delay line (ODL) plays a crucial role in optical signal manipulation, facilitating essential functions in optical buffering, beamforming, optical phased arrays, quantum state control, and optical coherence tomography systems.^[4–8] Conventional optical delay technologies,^[9] such as fiber-based delay lines and thermo-optic tuners, have inherent limitations in tuning speed, stability, power consumption, and integration density. These limitations make them inadequate for next-generation applications that require compact integration, high-speed response, and wide-range continuous tunability. The lithium niobate (LN) photonic platform has emerged as a revolutionary solution, leveraging its exceptional electro-optic coefficients, ultralow propagation loss, and strong nonlinear optical properties.^[10,11] Recent breakthroughs in fabrication techniques, such as deep-ultraviolet lithography and electron-beam lithography, combined with reactive ion etching, further confirm its significance.^[12–19] Particularly, the photolithography-assisted chemo-mechanical etching (PLACE) technique has also enabled high-yield

production of low-loss, high-density thin-film lithium niobate (TFLN) photonic integrated circuits.^[20] This technique combines sub-micrometer-scale resolution, atomic-level sidewall smoothness, and high throughput, facilitating the realization of various TFLN devices such as micro-disk resonators, narrow-linewidth micro-lasers, optical amplifiers, meter-scale delay lines, and photonic neural networks.^[21–28]

Current integrated ODL structures are primarily categorized into two types: resonance-enhanced and non-resonant. The resonance effect of materials induces drastic dispersion variations at resonant wavelengths, that enable light signals to circulate multiple times within the resonator, this phenomenon increases group delay and is utilized in devices such as ring resonators,^[29,30] photonic crystals,^[31] as well as in Brillouin scattering filters.^[32] Although these devices can achieve substantial delay times through compact designs, they often come with a limited operational bandwidth due to their resonant characteristics. On the other hand, non-resonant structures generally rely on path-switching Mach-Zehnder interferometers (MZIs) or refractive index tuning.^[33–35] While discrete path-switching structures can achieve a wide range of delays, they lack continuous tunability. In addition, refractive index modulation in LN waveguides allows for

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fine-tuning, although it is limited in terms of tuning efficiency.^[36]

In this Letter, to overcome the limitations of existing ODL structures, we experimentally demonstrate a TFLN-based ODL that utilizes a continuously tunable configuration with compact integration.^[37–39] Similar MZI-based tunable couplers in Si/III-V photonics typically serve as final stages in discrete-switch ODL, trading continuous tunability for extended range. Unlike conventional approaches relying on electric-field-induced refractive index modulation, our design employs a modified MZI integrated with dual electro-optically tunable couplers. Through synchronized tuning of the coupling ratios, the optical power distribution between the unbalanced MZI arms is progressively shifted, enabling continuous delay tuning from 0 to 293 ps while maintaining power fluctuations within 0.3 dB.

The proposed continuously tunable ODL is schematically depicted in Fig. 1(a). Its structure comprises a 2×2 MZI with two voltage-controlled tunable couplers and unbalanced-arms configuration. Each tunable coupler, structured as a 2×2 MZI, enables precise tuning of the coupling ratio by applying voltages to the electrodes on

its dual arms. Simultaneously, a phase-tuning electrode integrated on the unbalanced MZI arm introduces a controllable phase difference between these two arms, thereby enabling dynamic operating frequency tuning of the ODL. The working principle can be derived using transfer matrix theory. Assuming identical coupling ratios for both couplers $K_1 = K_2 = K = \cos^2(\theta/2)$, a fixed arm-length difference ΔL , and a propagation loss coefficient α , the transfer matrices for the 2×2 MZI and the unbalanced waveguides are expressed as follows:

$$S_{\text{MZI}}(\theta) = \begin{pmatrix} \cos\left(\frac{\theta}{2}\right) & -\sin\left(\frac{\theta}{2}\right) \\ \sin\left(\frac{\theta}{2}\right) & \cos\left(\frac{\theta}{2}\right) \end{pmatrix}, \quad (1)$$

$$S_{\text{wgs}}(\Omega) = \begin{pmatrix} e^{-\alpha L} e^{i(\Omega+\varphi)} & 0 \\ 0 & 1 \end{pmatrix}, \quad (2)$$

where Ω represents the initial phase difference of the unbalanced MZI arms and the phase-tuning electrode introduces an additional phase shift φ . The overall transfer matrix of the device is derived from the cascaded matrices of the MZI and unbalanced waveguides,

$$S_{\text{DLs}}(\Omega) = S_{\text{MZI}}(\theta) \cdot S_{\text{wgs}}(\Omega) \cdot S_{\text{MZI}}(\theta) \\ = \begin{pmatrix} K e^{-\alpha \Delta L} e^{i(\Omega+\varphi)} - (1-K) & -\sqrt{K(1-K)} e^{-\alpha \Delta L} e^{i(\Omega+\varphi)} - \sqrt{K(1-K)} \\ \sqrt{K(1-K)} e^{-\alpha \Delta L} e^{i(\Omega+\varphi)} + \sqrt{K(1-K)} & -(1-K) e^{-\alpha \Delta L} e^{i(\Omega+\varphi)} + K \end{pmatrix}. \quad (3)$$

When the device operates in the “bar” state (i.e., input and output ports reside on the same side), the power transfer function $|H(\Omega)|^2$ and group delay τ are expressed as

$$|H(\Omega)|^2 = K^2 e^{-2\alpha \Delta L} + (1-K)^2 \\ - 2K(1-K) e^{-\alpha \Delta L} \cos(\Omega + \varphi), \quad (4)$$

$$\tau = \frac{\partial H(\Omega)}{\partial \omega} \\ = \frac{K^2 - K(1-K) e^{\alpha \Delta L} \cos(\Omega + \varphi)}{K^2 + (1-K)^2 e^{\alpha \Delta L} - 2K(1-K) e^{\alpha \Delta L} \cos(\Omega + \varphi)} \Delta T, \quad (5)$$

where ΔT represents the delay difference of the unbalanced MZI arms and $\Omega = \omega \Delta T$ with ω denoting the angular frequency of the incident light. Specifically, when total phase difference $\phi = \Omega + \varphi = (2N+1)\pi$ (where N is an integer), the group delay simplifies to

$$\tau = \frac{K}{(1 - e^{\alpha \Delta L})K + e^{\alpha \Delta L}} \Delta T. \quad (6)$$

This theoretically indicates a continuous tunable mechanism for group delay τ from 0 to ΔT through controlling of the coupling ratio K , under fixed waveguide propagation loss and arms-length difference. Significantly, under the ultra-low-loss waveguide conditions (i.e., $e^{\alpha \Delta L} \approx 1$), the group delay satisfies a linear dependence $\tau \approx K \Delta T$.

Figures 1(b) and 1(c) present the normalized transmission and group delay characteristics of the proposed ODL as functions of phase difference under lossless conditions, with symmetrically configured coupling ratios (i.e.,

$K_1 = K_2 = K$). Specially, the phase difference can be expressed as $\phi = 2\pi f \Delta T + \varphi$ that depends on incident light frequency f and tunable phase φ . At the coupling ratios $K = 0$ and $K = 1$, all input light is either coupled into the reference (shorter) or delay (longer) arms, resulting in frequency-insensitive state where power transmission and group delay remain constant. The balanced configuration ($K = 0.5$) exhibits the minimum 3 dB bandwidth and maximum extinction ratio (ER) at the phase of $2N\pi$. This operating state simultaneously enables phase-independent group delay tuning. For all other values of K , both the transmission and the group delay demonstrate periodic modulation with a free spectral range (FSR) of $1/\Delta T$. Notably, power transmission overlap is observed between the complementary coupling pairs ($K \leftrightarrow 1-K$) due to the ideal lossless conditions. Actually, lower propagation losses result in reduced discrepancies between the transmission curves in above cases. Optimal ODL functionality is achieved when operating at critical phase points $\phi = (2N+1)\pi$ (vertical dashed lines), where power transmission becomes coupling-ratio invariant while normalized group delay exhibits linear dependence with K .

Figure 2(a) illustrates the design layout of a continuously tunable ODL, featuring a 2×2 unbalanced MZI with arm-length difference ΔL . This MZI integrates two voltage-tunable couplers, each constructed by cascading a symmetric 2×2 multimode interferometer (MMI) with a balanced MZI. The configuration enables continuous tuning of the coupling ratio from 0 to 1 through active control. An electro-optic phase shifter is embedded in two arms to

compensate for the frequency-dependent phase difference, ensuring stable control of the operating frequency. To enhance device controllability, auxiliary 2×2 MMI-MZI couplers with identical designs are integrated into both delay paths of unbalanced arms. These secondary couplers (i.e.,

MZI 3 and MZI 4) provide real-time monitoring and feedback correction of the primary MMI-MZI's (i.e., MZI 1 and MZI 5) operating state, while an additional tunable MZI 2 maintains optimal power distribution between output ports 2 and 3.

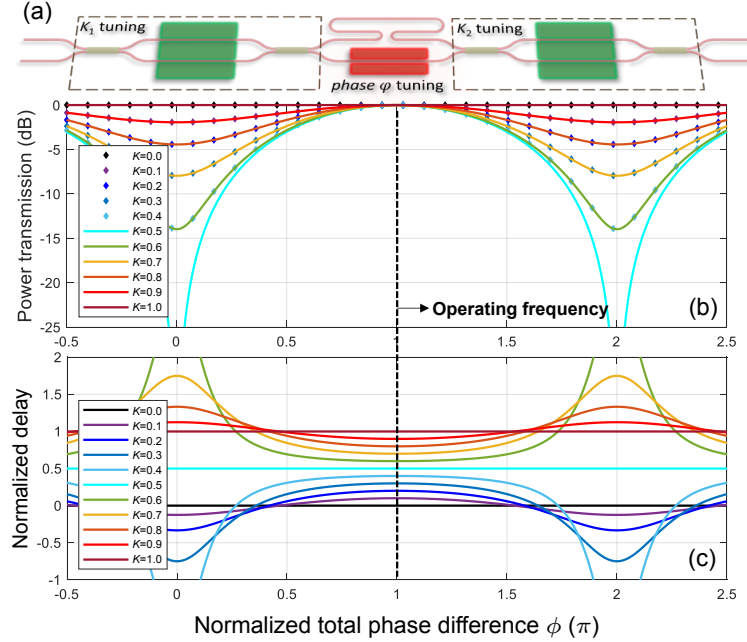


Fig. 1. (a) Schematic diagram of the proposed ODL based on an unbalanced MZI with two tunable couplers. (b) Normalized power transmission as the function of operating frequency. (c) Group delay versus phase difference under varying coupling ratios.

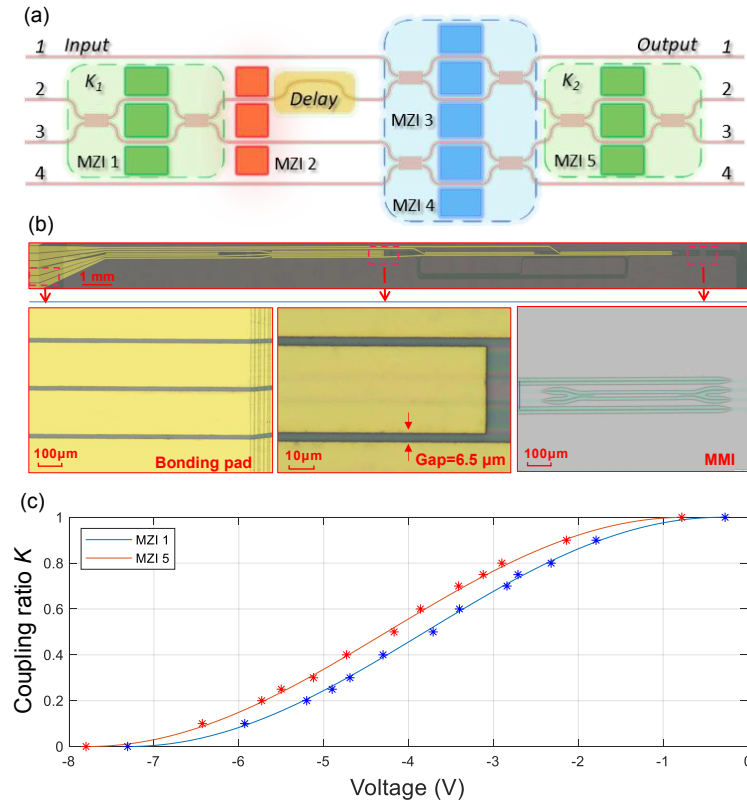


Fig. 2. (a) Design schematic of the fabricated ODL. (b) Micrographs of the ODL waveguides on TFLN with micro-electrodes. Insets show zoom-in micrographs of the bonding pad, electrode gap, and MMI structures. (c) Calibration curve between the applied electrical voltage and the achieved coupling ratio K of MZI 1 and MZI 5.

The proposed ODL is monolithically integrated on an X-cut TFLN platform, comprising a 500 nm LN layer bonded to a 4.7 μm silicon dioxide (SiO_2) buffer layer and a 500 μm silicon (Si) substrate. The ODL waveguide with 1 μm top width on TFLN is fabricated using the PLACE technique, more fabrication details can be found in Ref. [20]. For electrode integration, a multi-layer chromium (Cr)/gold (Au)/titanium (Ti) metallization stack (30/500/200 nm) is deposited via magnetron sputtering onto a 1.5 μm plasma-enhanced chemical vapor deposition SiO_2 cladding layer, serving dual purposes of optical isolation and electrical insulation between the electrodes and LN waveguides. The microfabrication process of the electrodes sequentially utilizes femtosecond laser ablation to define Ti masks, selective wet etching for structuring the Au layer, and the final removal of the sacrificial Cr adhesion layers and Ti masking layers, ultimately achieving pre-designed microelectrode patterns. Figure 2(b) shows top-view and zoomed-in micrographs of the fabricated continuously tunable ODL with an arm-length difference of 4 cm and corresponding microelectrodes integration. The device exhibits a compact footprint of 35×1.5 mm. The electrodes distributed on both sides of the waveguides are patterned with a 6.5 μm gap to optimize the electro-optic tuning efficiency. All electrodes are routed to the chip periphery, terminating in the 180 μm bonding pads for test compatibility.

Before conducting time-delay characterization tests, it is necessary to perform coupling ratio calibration for both MZI 1 and MZI 5. The calibration of coupling ratios can be concluded as follows. Firstly, we establish reference states for MZI 3(4) by coupling light into input port 1(4) and monitor the power of output ports 1, 2 and 3 (2, 3 and 4). The “bar” state is achieved when simultaneously maximizing output port 1(4) power while minimizing the combined power of output ports 2 and 3. Conversely, the “cross” state occurs when these power conditions reverse.

Secondly, with MZI 3 fixed in “cross” state and light injected through input port 1, we systematically calibrate MZI 5’s K_2 values by balancing output port 2 and port 3 powers. In other words, the coupling coefficient is defined as the normalized intensity ratio between output port 2 and the total power from ports 2 and 3. For MZI 1 calibration, dual cross-state conditions are implemented (i.e., MZI 3 and 4 both in “cross” state) with light entering input port 2, enabling K_1 value determination through power adjustments at output ports 1 and 4. The measured calibration K voltage curve of MZI 1 and MZI 5 is shown in Fig. 2(c), exhibiting a squared cosine functional dependency. Finally, with the light coupled into input port 2 and both MZI 3 and 4 stabilized in the “bar” state, we fix the wavelength at 1550 nm while ensuring MZI 1 and MZI 4 operate at identical K values. The device reaches its delay-tuning operation when output port 2 exhibits maximum power, at which point the applied bias voltage to MZI 2 corresponds to the operating point for the target wavelength, confirming that the phase difference in the ODL has been electro-optically tuned to the required $(2N+1)\pi$ phase shift.

Following the aforementioned calibration method, we experimentally characterize the normalized power transmission of ODL at coupling ratio $K = 0.5$ as a function of driving voltage applied to MZI 2. As shown in Fig. 3(a), the bar port (output port 2) and cross port (output port 3) transmission curves exhibit squared sine and squared cosine functional dependencies, respectively, showing a near-perfect symmetry and complementarity. To optimize group-delay tuning functionality, the operational voltage should be maintained at 8 V, where the bar-port output achieves its maximum value. The normalized transmission spectrum of the ODL at the $K = 0.5$ is further characterized in Fig. 3(b). The power transmission presents periodic wavelength modulation with a FSR of 27.2 pm, corresponding to a time delay of 293 ps. The result agrees

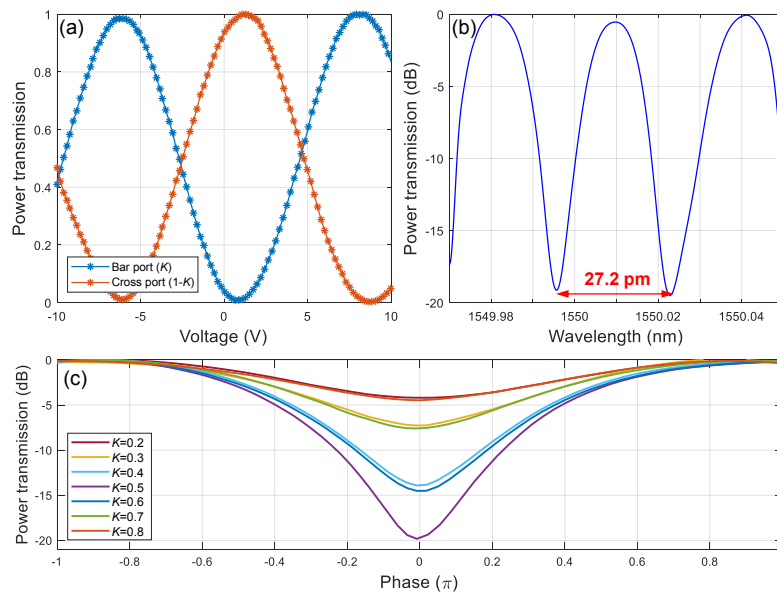


Fig. 3. Normalized power transmission characteristics of ODL, (a) Bar/cross port transmission versus driving voltage applied to MZI 2 at $K = 0.5$ exhibiting squared sine/cosine dependencies. (b) Wavelength-periodic transmission spectrum with 27.2 pm FSR at $K = 0.5$. (c) Phase-dependent transmission with coupling ratios tuning.

well with the theoretical prediction based on the fabricated waveguide's path difference of 4 cm and the group refractive index of 2.2 in the ridge LN waveguide. Figure 3(c) depicts the bar-port power transmission under phase tuning through periodic voltage scanning of the MZI 2 electrodes with varying coupling ratios from 0.2 to 0.8. At $K = 0.5$, the ER reaches its maximum value of -20 dB at zero phase. As the K values increase within $0 < K < 0.5$ or decrease within $0.5 < K < 1$, the ER progressively increases. Notably, smaller power transmission discrepancies (< 0.6 dB) are observed between the complementary coupling pairs ($K \leftrightarrow 1 - K$), demonstrating the low propagation loss waveguide (< 0.08 dB/cm) achieved through the PLACE fabrication technique. These experimental results agree well with the numerical model predictions outlined above.

To quantitatively visualize the time delay variation dynamics of the ODL, we implemented a time-calibrated experimental setup for real-time signal monitoring. As illustrated in Fig. 4(a), the measurement process proceeded as follows: A 1550 nm wavelength continuous light from a tunable laser source (CTL 1550, TOPTICA Photonics Inc.) was firstly polarization-controlled using a 3-paddle fiber polarization controller (FPC561, Thorlabs Inc.) before entering a commercialized LN modulator (MXAN-LN-20, Exail Photonics Inc.). This modulator was driven by 1-GSPS non-return-to-zero (NRZ) digital signals generated by an arbitrary waveform generator (AWG5200, Tektronix Inc.). The modulated optical signal was subsequently coupled into the input port of the device and then coupled out to the fiber, converted to electrical signal using a high-frequency photodetector (DXM30AF, Thorlabs Inc.) with the temporal waveform

evolution recorded on a real-time oscilloscope (MSO64B, Tektronix Inc.) with 1 GHz bandwidth. The optical carrier clock undergoes phase-locked synchronization to the AWG's timing source, with the AWG-sourced reference signal simultaneously routed to the oscilloscope for real-time monitoring of temporal waveform evolution.

Figure 4(b) illustrates the temporal waveform of original input signal and the modulated optical signals under varying K values. The original NRZ square waveform applied to the modulator exhibits discernible distortion after propagating through the photonic device, although the temporal displacement remains clearly observable. The intrinsic bandwidth-delay trade-off restricts high-fidelity operation to signals with bandwidths < 1.7 GHz at full delay in theory. This aligns with the distortion observed in Fig. 4(b), where the input NRZ signal bandwidth is comparable to the ODL's reliable bandwidth. In particular, the bandwidth is defined as the difference between two frequencies around the operating frequency where the delay drops or increases by 5%. In order to demonstrate the time-delay tunability, we measure signal waveform sampling at the discrete intervals of $\Delta K = 0.25$. Through half-maximum normalization of the intensity-time profile, five distinct delay states were quantified. We define $K = 0$ condition as the zero-delay reference baseline, the incremental relative delays demonstrate a linear dependence with increasing K values, as quantitatively verified by the slope of 293 ps shown in Fig. 4(c). Remarkably, the device maintains exceptional optical intensity stability during dynamic delay tuning, with intensity fluctuations of less than 0.3 dB, fulfilling the requirements of photonic applications that require wide-range temporal control.

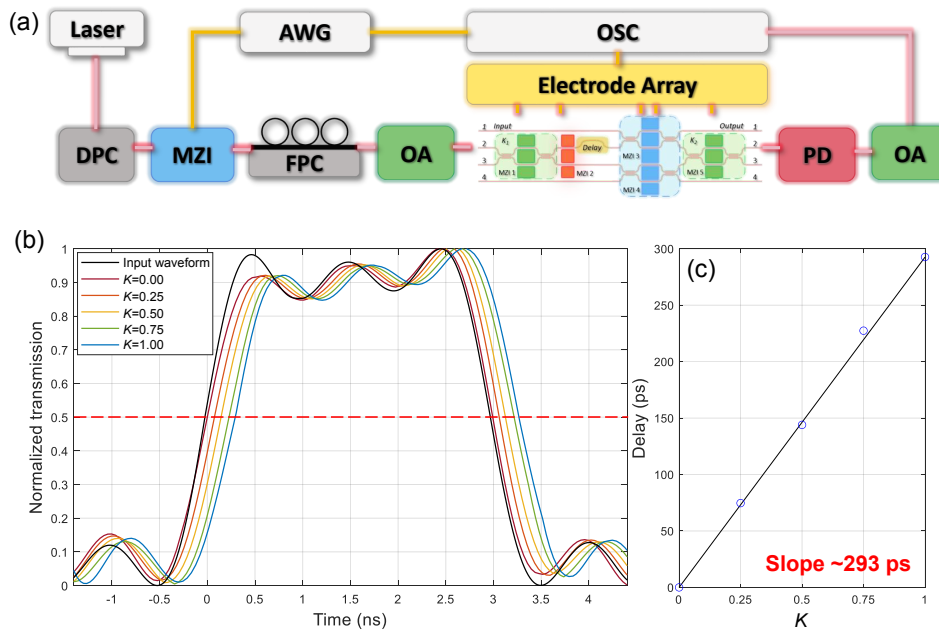


Fig. 4. (a) Schematic diagram of the group delay measurement setup. OA: optical amplifier; DPC: digital polarization controller; FPC: fiber polarization controller; AWG: arbitrary waveform generator; PD: photodetector; and OSC: Oscilloscope. (b) Normalized transmission of the temporal waveforms versus coupling ratios. The black line shows the original input waveform. (c) The delay shift of modulated light with different K values, showing a linear dependence with the slope of 293 ps.

In summary, we experimentally demonstrate a compact TFLN ODL (35×1.5 mm footprint) featuring wide-range continuous tunability. The device architecture employs an unbalanced MZI integrated dual tunable couplers, each comprising a 2×2 MMI and a balanced MZI structure. Through synchronous adjustment of the coupling coefficients, continuous delay tuning from 0 to 293 ps is achieved, corresponding to a 4 cm path-length difference between the MZI arms, demonstrating a wide continuous tuning range with remarkable intensity stability within 0.3 dB fluctuation. Our design prioritizes continuous tuning within a single stage. Future work may integrate cascaded stages on TFLN to combine wide range, large bandwidth and continuity. Such performance advantages position this platform as particularly promising for emerging applications in coherent optical communications, all-optical signal processing, quantum state manipulation, and phased array beamforming systems, where both wide-range delay control, power stability, and device miniaturization are critical requirements.

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References

- [1] Boes A, Corcoran B, Chang L, Bowers J, and Mitchell A 2018 *Laser Photon. Rev.* **12** 1700256
- [2] Marpaung D, Roeloffzen C, Heideman R, Leinse A, Sales S, and Capmany J 2013 *Laser Photon. Rev.* **7** 506
- [3] Elshaari A, Pernice W, Srinivasan K, Benson O, and Zwiller V 2020 *Nature Photon.* **14** 285
- [4] Shi S Q, Niu H S, Shi W H, Lin D D, Li S M, Pan S L, and Yun B F 2025 *Laser Photon. Rev.* **19** 2400663
- [5] Moralis-Pegios M, Mourgias-Alexandris G, Terzenidis N, Cherchi M, Harjanne M, Aalto T, Miliou A, Pleros N, and Vrysokinos K 2017 *IEEE Photon. Technol. Lett.* **30** 31
- [6] Liu Y, Isaac B, Kalkavage J, Adles E, Clark T, and Klamkin J 2019 *45th European Conference on Optical Communication (ECOC)* **1**
- [7] Yang D H and Lin W P 2015 *Opt. Commun.* **350** 90
- [8] Huang D, Swanson E A, Lin C P, Schuman J S, Stinson W G, Chang W, Hee M R, Flotte T, Gregory K, Puliafito C A, and Fujimoto J G 2019 *Science* **254** 1178
- [9] Ding M, Feng Z T, Marpaung D, Zhang X, Komanec M, Suslov D, Dousek D, Zvánovec S, Fokoua E R N, Bradley T D, Poletti F, Richardson D J, and Slavík R 2021 *J. Lightwave Technol.* **39** 2311
- [10] Turner E 1966 *Appl. Phys. Lett.* **8** 303
- [11] Prokhorov A and Kuzminov Y 1990 *Physics and Chemistry of Crystalline Lithium Niobate* (Adam Hilger)
- [12] Wang J, Bo F, Wan S, Li W X, Gao F, Li J J, Zhang G Q, and Xu J J 2015 *Opt. Express* **23** 23072
- [13] Ye Z L, Li S F, Cui G X, Yin Z J, Wang X B, Zhao G, Hu X O, and Zhu S N 2024 *J. Synth. Cryst.* **53** 426
- [14] Zhang M, Wang C, Cheng R, Shams-Ansari A, and Lončar M 2017 *Optica* **4** 1536
- [15] Luke K, Kharel P, Reimer C, He L Y, Loncar M, and Zhang M 2020 *Opt. Express* **28** 24452
- [16] Wang C, Burek M J, Lin Z, Atikian H A, Venkataraman V, Huang I, Stark P, and Lončar M 2014 *Opt. Express* **22** 30924
- [17] Zhang K, Chen Z X, Feng H K, Wong W H, Pun E Y, and Wang C 2021 *Chin. Opt. Lett.* **19** 060010
- [18] He M B, Xu M Y, Ren Y X, Jian J, Ruan Z L, Xu Y S, Gao S Q, Sun S H, Wen X Q, Zhou L D, Guo C J, Chen H, Yu S Y, Liu L, and Cai X L 2019 *Nat. Photon.* **13** 359
- [19] Zhuang R J, He J Z, Qi Y F, and Li Y 2023 *Adv. Mater.* **35** 2208113
- [20] Wu R B, Wang M, Xu J, Qi J, Chu W, Fang Z W, Zhang J H, Zhou J X, Qiao L L, Chai Z F, Lin J T, and Cheng Y 2018 *Nanomaterials* **8** 910
- [21] Wu R B, Zhang J H, Yao N, Fang W, Qiao L L, Chai Z F, and Cheng Y 2018 *Opt. Lett.* **43** 4116
- [22] Gao R H, Zhang H S, Bo F, Fang W, Hao Z Z, Yao N, Lin J T, Guan J L, Deng L, Wang M, Qiao L, and Cheng Y 2021 *New J. Phys.* **23** 123027
- [23] Wu R B, Lin J T, Wang M, Fang Z W, Chu W, Zhang J H, Zhou J X, and Cheng Y 2019 *Opt. Lett.* **44** 4698
- [24] Zhou J X, Liang Y T, Liu Z X, Chu W, Zhang H, Yin D F, Fang Z W, Wu R B, Zhang J H, Chen W, Wang Z, Zhou Y, Wang M, and Cheng Y 2021 *Laser Photon. Rev.* **15** 2100030
- [25] Liang Y T, Zhou J X, Liu Z, Zhang H, Fang Z W, Zhou Y, Yin D F, Lin J T, Yu J P, Wu R B, Wang M, and Cheng Y 2022 *Nanophotonics* **11** 1033
- [26] Zhou J X, Huang T, Fang Z W, Wu R B, Zhou Y, Liu J, Zhang H S, Wang Z, Wang M, and Cheng Y 2022 *Opt. Lett.* **47** 5599
- [27] Zhou J X, Gao R H, Lin J, Wang M, Chu W, Li W B, Yin D F, Deng L, Fang Z W, Zhang J H, Wu R B, and Cheng Y 2022 *Chin. Phys. Lett.* **37** 084201
- [28] Zheng Y, Wu R B, Ren Y, Bao R, Liu J, Ma Y, Wang M, and Cheng Y 2024 *Laser Photon. Rev.* **18** 2400565
- [29] Cardenas J, Foster M A, Sherwood-Droz N, Poitras C B, Lira H L, Zhang B, Gaeta A L, Khurgin J B, Morton P, and Lipson M 2010 *Opt. Express* **18** 26525
- [30] Tejaswini B R and Chandrika T N 2019 *1st International Conference on Advances in Information Technology (ICAIT)* p. 512
- [31] Melloni A, Canciamilla A, Ferrari C, Morichetti F, O'Faolain L, Krauss T F, Rue R D L, Samarelli A, and Sorel M 2010 *IEEE Photon.* **2** 181
- [32] Marpaung D, Morrison B, Pagani M, Pant R, Choi D Y, Luther-Davies B, Madden S J, and Eggleton B J 2015 *Optica* **2** 76
- [33] Wang X Y, Zhou L J, Li R F, Xie J Y, Lu L J, Wu K, and Chen J P 2017 *Optica* **4** 507
- [34] Wang Y F, Hu J L, Liang W, Wang Q, Ke W, Yu S Y, Lin Z J, and Cai X L 2024 *50th European Conference on Optical Communication (ECOC)* p. 710
- [35] Ke W, Lin Y M, He M B, Xu M Y, Zhang J X, Lin Z J, Yu S Y, and Cai X L 2022 *Photon. Res.* **10** 2575
- [36] Song L B, Chen J M, Wu R B, Zheng Y, Liu Z X, Wang G H, Sun C, Wang M, and Cheng Y 2023 *Opt. Lett.* **48** 2261
- [37] Melati D, Waqas A, Mushtaq Z, and Melloni A 2018 *IEEE J. Sel. Top. Quantum Electron.* **24** 1
- [38] Song Q Q 2022 *IEEE Photon. J.* **4** 1
- [39] Waqas A, Khani T K, and Corbett B 2024 *IEEE Silicon Photonics Conference (SiPhotonics)* p. 1